Example 7.21 The volume of a solid hemisphere is $29106\ \text{cm}^3$. Another hemisphere whose volume is two-third of the above is carved out. Find the radius of the new hemisphere.

Solution Let r be the radius of the hemisphere.

Given that, volume of the hemisphere = $29106\ \text{cm}^3$

$$\text{Now, volume of new hemisphere} = \frac{2}{3}(\text{Volume of original sphere})$$

$$= \frac{2}{3} \times 29106$$

$$\text{Volume of new hemisphere} = 19404\ \text{cm}^3$$

$$\frac{2}{3}\pi r^3 = 19404$$

$$r^3 = \frac{19404 \times 3 \times 7}{2 \times 22} = 9261$$

$$r = \sqrt[3]{9261} = 21\ \text{cm}$$

Therefore, $r = 21\ \text{cm}$

Thinking Corner

1. Give any two real life examples of sphere and hemisphere.
2. A plane along a great circle will split the sphere into _____ parts.
3. If the volume and surface area of a sphere are numerically equal, then the radius of the sphere is _____.

Example 7.22 Calculate the mass of a hollow brass sphere if the inner diameter is 14 cm and thickness is 1mm, and whose density is $17.3\ \text{g/ cm}^3$. (**Hint:** *mass = density × volume*)

Solution Let r and R be the inner and outer radii of the hollow sphere.

Given that, inner diameter $d = 14$ cm; inner radius $r = 7$ cm; thickness $= 1$ mm $= \frac{1}{10}$ cm

$$\text{Outer radius } R = 7 + \frac{1}{10} = \frac{71}{10} = 7.1\ \text{cm}$$

$$\text{Volume of hollow sphere} = \frac{4}{3}\pi\left(R^3 - r^3\right)\ \text{cu. units}$$

$$= \frac{4}{3} \times \frac{22}{7}(357.91 - 343) = 62.48\ \text{cm}^3$$

But, density of brass in $1\ \text{cm}^3 = 17.3$ gm

$$\text{Total mass} = 17.3 \times 62.48 = 1080.90\ \text{gm}$$

Therefore, total mass is 1080.90 grams.

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